Risk Assessment

Prepared By: Antonio Mendez

Role: Cybersecurity Analyst (Simulation)

Organization: RetailNova

Classification: Confidential

Date: May 27, 2026, 9.49pm CDT

Executive Summary:

RetailNova has experienced three security incidents to date. The initial breach occurred on October 12, 2023, resulting in a credential compromise triggered by a targeted phishing campaign against customer service representatives.

The second incident, a ransomware attempt on December 2, 2024, was successfully mitigated by endpoint protection systems. However, the remediation process needed a two-hour operational downtime.

The most recent incident occurred on February 20, 2025, involving a data leak from a third-party marketing vendor that exposed the email addresses of approximately 5,000 customers.

Incident Summary:

Incident	Date Detected	Root Cause	Data/Systems Affected	Risk Level
Breach 1: Credential Compromise	October 12, 2023	Phishing / Social Engineering	Customer Service Representatives	Critical
Breach 2: Ransomware Attack	December 2, 2024	Malware Exploitation	Corporate Network (POS, RBAC, VPN)	High
Breach 3: Vendor Data Leak	February 20, 2025	Third-Party Security Breach	Customers Email Addresses (5,000 records)	Medium

Incident Analysis and Findings:

The incidents outlined above demonstrate a pattern of cybersecurity weaknesses affecting employee security awareness, system protection, and third-party risk management. Each event had a different attack vector and business impact, highlighting the importance of a layered security strategy.

Credential Compromise (October 2023):

The first incident involved a phishing campaign targeting customer service representatives. Attackers used fraudulent emails designed to impersonate company leadership to obtain employee credentials and facilitate unauthorized financial transactions. As a result, employee accounts were compromised, leading to financial losses and increased organizational risk.

To reduce the likelihood of similar incidents, RetailNova should implement Multi-Factor Authentication (MFA), advanced email filtering, URL scanning, and regular phishing awareness training. Maintaining up-to-date operating systems and software is also critical to reducing vulnerabilities that attackers may exploit.

Ransomware Attempt (December 2024):

The second incident involved a ransomware attack against RetailNova's internal network. Although endpoint protection successfully prevented file encryption, attackers were able to gain access to internal systems, resulting in approximately two hours of operational downtime.

This disruption affected business operations, reduced service availability, and potentially impacted revenue and customer trust. To strengthen resilience against future ransomware attacks, RetailNova should implement network segmentation, isolate critical systems such as POS terminals and inventory servers, and enhance endpoint detection and response

capabilities. Regular backup testing and incident response exercises should also be conducted.

Third-Party Vendor Data Leak (February 2025):

The third incident originated from a security breach involving a third-party marketing vendor. Approximately 5,000 customer email addresses were exposed, increasing the risk of phishing campaigns and unauthorized access to personal information.

To mitigate third-party risks, RetailNova should strengthen vendor risk management processes, perform periodic security assessments of third-party providers, and ensure sensitive customer information is encrypted at rest. Data tokenization and clear breach notification procedures should also be implemented to improve customer protection and regulatory compliance.

Overall Risk Assessment:

The analysis indicates that RetailNova faces elevated cybersecurity risks due to social engineering attacks, operational vulnerabilities, and reliance on third-party vendors. The recurring nature of these incidents suggests that existing security controls were insufficient to prevent or effectively mitigate emerging threats.

Conclusion:

Over the three-year assessment period, RetailNova experienced multiple cybersecurity incidents that affected confidentiality, availability, and customer trust. While remediation efforts have reduced overall exposure, residual risk remains due to evolving cyber threats and the possibility of human error. Continuous monitoring, employee education, and ongoing improvements to security controls will be essential to maintaining a strong cybersecurity posture and supporting long-term business resilience.